

Travaux de Shimura

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Let X/Γ be the quotient of a hermitian symmetric domain X by a discrete arithmetic group Γ (assumed to be defined by congruence conditions, see 1.7). According to Baily and Borel, X/Γ is a complex algebraic variety. In the articles [10][11][12][13][14], Shimura shows among others, for many of those varieties, that they can be defined over explicit number fields. This is the subject of this paper.

To obtain proper statements, one is forced to use an adelic language. This leads, for a given "congruence condition", to considering a scheme over $\mathbb{C}$, a finite disjoint union of varieties of type X/Γ (see §§1 and 2). In many cases, this scheme can be defined over a number field E , independent of the congruence condition considered. Its geometric connected components, however, will be defined over class fields of E . This phenomenon, although essential, will be neglected in the remainder of this introduction.

In a small number of cases, X/Γ can be interpreted as the set of isomorphism classes of complex abelian varieties, equipped with some additional algebraic structures (polarizations, endomorphisms, structures on points of order n). The solution, over $\mathbb{Q}$, of a corresponding moduli problem then provides a scheme M over an explicit number field F , for which X/Γ is the set of complex points. Such an M will be called a "model" of X/Γ . The fundamental case is that of principally polarized abelian varieties, equipped with a level n structure (1.6, 1.11, 4.16, 4.17 and 4.21).

In other cases, X/Γ can be interpreted as the set of isomorphism classes of complex abelian varieties A equipped with some structures as above, and with some integral cohomology classes of type (p, p) in $H^{2p}(A \times \cdots \times A, \mathbb{C})$ (cf. Mumford [7]). In this case, the idea will be to construct a model M of X/Γ as a subscheme of a model M' already constructed for a variety X'/Γ' . Roughly speaking, one begins by constructing in M' the points corresponding to abelian varieties of C.M. type (= maximal complex multiplication). One then defines M as the closure of the set of these points. This construction relies on the detailed knowledge of abelian varieties of C.M. type provided by [16].

In the cases considered above, one gets a great deal of information about the fields of definition of the points of M corresponding to abelian varieties of C.M. type. This allows one to give partial solutions to Hilbert's 12th problem (cf. the introduction of [15] and 3.15, 3.16).

By examining what these cases have in common, one arrives at the notion of "canonical model" (§3). It is shown that there is at most one "canonical model" of the X/Γ (5.5). Descent techniques then allow the construction of some canonical models that seem far from being solutions to moduli problems ([12][13] [14] and [5]). For the description of these "strange models", one is referred to §§5 and 6.

We denote by $\mathbb{Z}/n(1)$ both the group scheme μ_n of n -th roots of unity and the group of n -th roots of unity in a fixed algebraically closed field k . For $k = \mathbb{C}$, the exponential map allows us to identify $\mathbb{Z}/n(1)$ with $\mathbb{Z}/n$. The $\mathbb{Z}/n(1)$ form a projective system: the transition maps are $\varphi_{n,m} : x \mapsto x^{n/m}$.

We set

$$\hat{\mathbb{Z}} = \varprojlim \mathbb{Z}/n\mathbb{Z} \xrightarrow{\sim} \prod \mathbb{Z}_p, \quad \hat{\mathbb{Z}}(1) = \varprojlim \mathbb{Z}/n\mathbb{Z}(1),$$

$$\mathbb{A}^f = \mathbb{Q} \otimes \hat{\mathbb{Z}} \xrightarrow{\sim} \prod' \mathbb{Q}_p \quad (\text{restricted product}), \quad \mathbb{A}^f(1) = \mathbb{Q} \otimes \hat{\mathbb{Z}}(1) = \mathbb{A}^f \otimes_{\hat{\mathbb{Z}}} \hat{\mathbb{Z}}(1)$$

and we denote by $\mathbb{A}$ the ring of adèles of $\mathbb{Q}$.

We will use the following notations:

$\pi_0(X)$ (for X a topological space): the set of connected components of X . In the following, $\pi_0(X)$, equipped with the quotient topology of X , will be discrete or compact totally disconnected.

G^0 : connected component of the identity of a topological group G (or of a pointed topological space).

$G(K), G_K, G \otimes_F K$: for G a scheme over F (for example, an algebraic group over F) and K an F -algebra, we denote by $G(K)$ the set of K -valued points of G and by G_K or $G \otimes_F K$ the scheme over K obtained from G by base change.

E^* : for E an algebra of finite rank over a field F , E^* denotes the algebraic group over F of invertible elements of E . For E a number field and $F = \mathbb{Q}$, $E^*(\mathbb{A})/E^*(\mathbb{Q})$ is the idèle class group of E .

We say that an algebraic group G over $\mathbb{Q}$ satisfies the Hasse principle if $H^1(\mathbb{Q}, G) =_{\text{def}} H^1(\text{Gal}(\overline{\mathbb{Q}}/\mathbb{Q}), G(\overline{\mathbb{Q}}))$ injects into the product over all places of $H^1(\mathbb{Q}_v, G_{\mathbb{Q}_v})$.

We will use the following theorems.

(0.1) Hasse principle: a semi-simple simply connected group, without factor of type E_8 , satisfies the Hasse principle [3].

(0.2) A semi-simple simply connected group G over a non-archimedean local field K satisfies $H^1(K, G) = 0$ (a proof independent of the classification is announced in Bruhat-Tits [1]).

(0.3) Strong approximation theorem: let G be a semi-simple simply connected group over a global field K . If G is K -simple, and if v is a place of K such that $G(K_v)$ is non-compact, then $G(K_v)G(K)$ is dense in $G(\mathbb{A})$ (for a proof independent of the classification, see Platonov [9]).

(0.4) Real approximation theorem: let G be a connected (linear) algebraic group over $\mathbb{Q}$. Then $G(\mathbb{Q})$ is dense in $G(\mathbb{R})$ (to prove it, one easily reduces to the case of tori, which is a corollary of ([4] 5.1)).

Chapter 1

The adelic language

1.1

Let G be a reductive group over $\mathbb{Q}$. We denote by G' the derived group of the neutral component, by C the center of this neutral component, and set $T = G/G'$. We have exact sequences of algebraic groups

$$(1.1.1) \quad 0 \rightarrow C \rightarrow G \rightarrow G/C \rightarrow 0,$$

$$(1.1.2) \quad 0 \rightarrow G' \rightarrow G \xrightarrow{\nu} T \rightarrow 0$$

and the canonical maps from C to T and from G' to G/C are isogenies with kernel $C \cap G'$.

1.2

Let $\mathbb{S}$ be the real algebraic group of invertible elements of the $\mathbb{R}$ -algebra $\mathbb{C}$. It is also the real algebraic group obtained by Weil restriction of scalars from $\mathbb{C}$ to $\mathbb{R}$ of the multiplicative group $\mathbb{G}_m$. We have $\mathbb{S}(\mathbb{R}) = \mathbb{C}^*$, and $\mathbb{S}_{\mathbb{C}} \simeq \mathbb{G}_{m,\mathbb{C}} \times \mathbb{G}_{m,\mathbb{C}}$. The group $\text{Hom}(\mathbb{S}_{\mathbb{C}}, \mathbb{G}_{m,\mathbb{C}})$ of characters of $\mathbb{S}_{\mathbb{C}}$ has basis the characters z and $\bar{z}$ such that the composite

$$\mathbb{C}^* = \mathbb{S}(\mathbb{R}) \hookrightarrow \mathbb{S}(\mathbb{C}) \xrightarrow{z, \bar{z}} \mathbb{C}^*$$

is respectively the identity and complex conjugation.

Let V be a real vector space. For any representation $h : \mathbb{S} \rightarrow \text{GL}(V)$, we denote by V^{pq} the subspace of $V_{\mathbb{C}}$ on which $\mathbb{S}$ acts by the character $z^p \bar{z}^q$. The V^{pq} form a bigradation of $V_{\mathbb{C}}$, and the construction $h \mapsto V^{pq}$ identifies the representations of $\mathbb{S}$ in $\text{GL}(V)$ with the Hodge bigradations of $V_{\mathbb{C}}$, i.e., with the bigradations of $V_{\mathbb{C}}$ such that $V^{pq} = \overline{V^{qp}}$. We denote by $F(h)$ the Hodge filtration of $V_{\mathbb{C}}$:

$$F(h)^p = \bigoplus_{p' \geq p} V^{p'q'}.$$

1.3

We denote by $r : \mathbb{G}_{m,\mathbb{C}} \rightarrow \mathbb{S}_{\mathbb{C}}$ the $\mathbb{C}$ -homomorphism such that $(z^p \bar{z}^q) \circ r = (x \mapsto x^p)$, and by $w : \mathbb{G}_{m,\mathbb{R}} \rightarrow \mathbb{S}$ the $\mathbb{R}$ -homomorphism such that $(z^p \bar{z}^q) \circ w = (x \mapsto x^{p+q})$. For a representation $h : \mathbb{S} \rightarrow \mathrm{GL}(V)$ and $v^{pq} \in V^{pq}$, we have

$$\begin{aligned} hr(x) \cdot v^{pq} &= x^p \cdot v^{pq} & \text{and} \\ hw(x) \cdot v^{pq} &= x^{p+q} \cdot v^{pq}. \end{aligned}$$

The composite $w : \mathbb{R}^* = \mathbb{G}_m(\mathbb{R}) \xrightarrow{w} \mathbb{S}(\mathbb{R}) = \mathbb{C}^*$ is the obvious inclusion.

1.4

Let $h : \mathbb{S} \rightarrow G_{\mathbb{R}}$ be a homomorphism. For any representation $\rho : G \rightarrow \mathrm{GL}(V)$ of G on a $\mathbb{Q}$ -vector space V , ρh defines a bigradation of $V_{\mathbb{C}}$ (1.2). If V is faithful, and if G is the subgroup of $\mathrm{GL}(V)$ fixing some tensors s_i , the construction $h \mapsto (V^{pq})$ identifies the homomorphisms h with the Hodge bigradations of $V_{\mathbb{C}}$ such that the s_i are of type $(0,0)$.

1.5

Let $h_0 : \mathbb{S} \rightarrow G_{\mathbb{R}}$ be a homomorphism satisfying the following conditions (cf. [7] and [2]; (1.5.2) is motivated by Griffiths' transversality theorem).

- (1.5.1) The image of $h_0 w : \mathbb{G}_m \rightarrow G_{\mathbb{R}}$ is central. We call $h_0 w$ the weight of h_0 ;
- (1.5.2) the Hodge bigradation of the complexified Lie algebra $\mathrm{Lie}(G)_{\mathbb{C}}$ (for the adjoint representation) (1.2) is of type $(-1,1) + (0,0) + (1,-1)$;
- (1.5.3) the automorphism $\mathrm{ad} h_0(i)$ of G is an involution (by (1.5.1)) and induces a Cartan involution of G' .

Let K_{∞} be the centralizer of h_0 in $G(\mathbb{R})$; it contains the centre of $G(\mathbb{R})$, and $K_{\infty} \cap G'(\mathbb{R})^0$ is a maximal compact subgroup of $G'(\mathbb{R})^0$, identical to the centralizer in $G'(\mathbb{R})^0$ of $h_0(i)$. The space $K_{\infty} \backslash G(\mathbb{R})$ is identified with the set X of conjugates of h_0 by an element of $G(\mathbb{R})$. One checks that it possesses one and only one complex structure such that, for every representation $\rho : G \rightarrow \mathrm{GL}(V)$, the filtration $F(\rho h)$ of $V_{\mathbb{C}}$ depends holomorphically on $h \in X$. This structure is right-invariant under $G(\mathbb{R})$, and the connected components of X are Hermitian symmetric domains. We denote by X^0 the connected component of h_0 .

1.6

Example I. Let $\mathrm{Gp}(V)$ be the group of symplectic similitudes of a real vector space V equipped with a non-degenerate alternating form ψ . Then there exists one and only one conjugacy class X of homomorphisms $h : \mathbb{S} \rightarrow \mathrm{Gp}(V)$ satisfying conditions (1.5.1) and of weight -1 , i.e. with weight $hw : \mathbb{G}_m \rightarrow \mathrm{Gp}(V)$ given by $x \mapsto$ homothety of ratio x^{-1} . The space X is identified with the union of two Siegel half-spaces.

The construction 1.5 identifies the $h \in X$ with decompositions $V_{\mathbb{C}} = V^{-1,0} \oplus V^{0,-1}$ of $V_{\mathbb{C}} = V \otimes_{\mathbb{R}} \mathbb{C}$ such that $V^{pq} = \overline{V^{qp}}$, $V^{0,-1}$ is totally isotropic for ψ , and, if C denotes the endomorphism of V which induces multiplication by i^{p-q} on V^{pq} , the symmetric form $\psi(x, Cy) = \psi(x, h(i)y)$ on V is definite (> 0 or < 0).

1.7

Let Γ be an arithmetic subgroup of $G(\mathbb{Q})$. We are interested in the case where Γ is defined by congruence conditions. If $V_{\mathbb{Z}}$ is an integral lattice in a faithful representation V of G , this means that there exists n such that Γ contains, with finite index, the group $\Gamma_n \subset G(\mathbb{Q})$ of those $\gamma \in G(\mathbb{Q})$ such that $\gamma \in G(\mathbb{R})^0$, $\gamma V_{\mathbb{Z}} = V_{\mathbb{Z}}$ and $\gamma \equiv \text{Id} \pmod{n}$. According to Baily and Borel, the quotient X^0/Γ is naturally endowed with a structure of quasi-projective scheme over $\mathbb{C}$. Let K_n be the compact open subgroup of $G(\mathbb{A}^f)$ consisting of those $k = (k_p)$ such that $k_p \cdot V_{\mathbb{Z}} \otimes \mathbb{Z}_p = V_{\mathbb{Z}} \otimes \mathbb{Z}_p$ and $k_p \equiv 1 \pmod{n}$ for $p \mid n$. Then Γ_n is the intersection, in $G(\mathbb{A})$, of $G(\mathbb{Q})$ with $G(\mathbb{R})^0 \times K_n$.

1.8

Let K be a compact open subgroup of $G(\mathbb{A}^f)$. The space

$$K_{\infty} \times K \backslash G(\mathbb{A})/G(\mathbb{Q}) = K \backslash X \times G(\mathbb{A}^f)/G(\mathbb{Q}) = X \times (K \backslash G(\mathbb{A}^f))/G(\mathbb{Q})$$

is then a finite sum of spaces of the type considered in 1.7. It is therefore a quasi-projective scheme over $\mathbb{C}$. We denote it by ${}_K M_{\mathbb{C}}(G, h_0)$ (or simply ${}_K M_{\mathbb{C}}(G)$ or ${}_K M_{\mathbb{C}}$).

As K becomes smaller, these schemes form a projective system of schemes with finite and surjective transition morphisms

$$(1.8.1) \quad {}_K M_{\mathbb{C}}(G, h_0) \quad (K \rightarrow 0).$$

We denote by $M_{\mathbb{C}}(G, h_0)$ (or simply $M_{\mathbb{C}}(G)$, or $M_{\mathbb{C}}$) the quasi-compact and separated scheme (not of finite type for $G \neq \{1\}$) which is the projective limit of (1.8.1)

$$(1.8.2) \quad M_{\mathbb{C}}(G, h_0) = \varprojlim {}_K M_{\mathbb{C}}(G, h_0).$$

We must regard it as an avatar of the projective system (1.8.1). The group $G(\mathbb{A}^f)$ acts on $M_{\mathbb{C}}$, or, if one prefers, on the pro-object defined by the projective system (1.8.1), but of course not on the individual ${}_K M_{\mathbb{C}}$ (cf. 3.2). We have

$$K \backslash M_{\mathbb{C}} = {}_K M_{\mathbb{C}},$$

so that

$$(1.8.3) \quad M_{\mathbb{C}} = \varprojlim K \backslash M_{\mathbb{C}} \quad (K \text{ compact open in } G(\mathbb{A}^f)).$$

This is expressed by saying that $G(\mathbb{A}^f)$ acts continuously on $M_{\mathbb{C}}$.

The following construction will be useful for interpreting the ${}_K M_{\mathbb{C}}$ as moduli schemes.

1.9

Let H be an algebraic group over $\mathbb{Q}$ and V a faithful representation of H . We define an H -structure $\bar{\iota}$ on a vector space W as an isomorphism of V with W , given modulo $H(\mathbb{Q})$: $\iota \in \text{Isom}(V, W)/H(\mathbb{Q})$. If $H(\mathbb{Q})$ is the group of automorphisms of a structure s of type Σ on V , then an H -structure on W is identified with a structure t of the same type on W , isomorphic to s (with $\bar{\iota} = \text{Isom}((V, s), (W, t))$). Let W be equipped with an H -structure $\bar{\iota}$. The algebraic group $\iota H \iota^{-1} \subset \text{GL}(W)$ ($\iota \in \bar{\iota}$) depends only on $\bar{\iota}$. We denote it by $H^{\bar{\iota}}$. Let A be a $\mathbb{Q}$ -algebra. The set $\text{Isom}_{\bar{\iota}}(W \otimes A, V \otimes A)$ of admissible isomorphisms of $W \otimes A$ with $V \otimes A$ is the set of isomorphisms k such that, for $\iota \in \bar{\iota}$, we have $k\iota \in H(A)$.

1.10

Let (G, h_0) be as in 1.5, V a faithful linear representation of G , K a compact open subgroup of $G(\mathbb{A}^f)$, and consider objects H of the following type: H consists of

- (a) a vector space $H_{\mathbb{Q}}$ over $\mathbb{Q}$, equipped with a G -structure $\bar{\iota}$ (1.8);
- (b) a homomorphism $h : \mathbb{S} \rightarrow G_{\mathbb{R}}^{\bar{\iota}}$ such that, for $\iota \in \bar{\iota}$, $\iota^{-1}h\iota : \mathbb{S} \rightarrow G_{\mathbb{R}}$ is conjugate to h_0 ;
- (c) a class $\bar{k} \in K \backslash \text{Isom}_{\bar{\iota}}(H \otimes \mathbb{A}^f, V \otimes \mathbb{A}^f)$ (1.9).

1.11

Example I (continued from 1.6). Let V be a $\mathbb{Q}$ -vector space equipped with a non-degenerate alternating form ψ , G the group of symplectic similitudes of V , $V_{\mathbb{Z}}$ a lattice on which ψ takes integer values with discriminant l , and h_0 as in 1.6. Take $K = \prod_{\ell} \text{Gp}(V_{\mathbb{Z}} \otimes \mathbb{Z}_{\ell})$.

A datum (a) is interpreted as a vector space H of the same dimension as V , equipped with an alternating form ψ given up to a rational factor. A datum (b) is interpreted as a Hodge bigradation of $H_{\mathbb{C}}$, of type 1.6. A datum (c) is interpreted as an integral lattice $H_{\mathbb{Z}}$ of H on which a rational multiple of ψ is integer-valued with discriminant l : $\bar{k}$ is the set of k such that, for every ℓ , $V_{\mathbb{Z}} \otimes \mathbb{Z}_{\ell} = k_{\ell}(H_{\mathbb{Z}} \otimes \mathbb{Z}_{\ell})$. We can normalize ψ by requiring it to be integer-valued of discriminant l on $H_{\mathbb{Z}}$, and to satisfy $\psi(x, Cx) > 0$ (cf. 1.6).

Let n be an integer, and K_n the subgroup of K consisting of those k such that $k \equiv 1 \pmod{n}$. For K_n , a datum (c) is interpreted as the lattice $H_{\mathbb{Z}}$ as above, together with a symplectic similitude $H_{\mathbb{Z}}/nH_{\mathbb{Z}} \xrightarrow{\sim} V_{\mathbb{Z}}/nV_{\mathbb{Z}}$.

1.12

Under the hypotheses of 1.10, the points of ${}_K M_{\mathbb{C}}(G, h_0)$ are in bijective correspondence with the isomorphism classes of objects H of type 1.10. Indeed, given $H = (H_{\mathbb{Q}}, \bar{\iota}, h, \bar{k})$ equipped with $\iota \in \bar{\iota}$, we associate to it the morphism $\iota^{-1}h\iota : \mathbb{S} \rightarrow G_{\mathbb{R}}$, an element of $X \simeq K_{\infty} \backslash G(\mathbb{R})$, and the class $\bar{k}\iota$ in $K \backslash G(\mathbb{A}^f)$, giving altogether an element of $K \backslash X \times G(\mathbb{A}^f)$; given H alone, we only associate an element of ${}_K M_{\mathbb{C}}(G, h_0) = K \backslash X \times G(\mathbb{A}^f)/G(\mathbb{Q}) = K_{\infty} \times K \backslash G(\mathbb{A})/G(\mathbb{Q})$, which uniquely determines the isomorphism class of H .

In some cases (see 4.11), giving an object H as above is equivalent to giving an abelian variety equipped with some additional structures. One then obtains an interpretation of ${}_K M_{\mathbb{C}}(G)$ (and of $M_{\mathbb{C}}(G)$) as the (coarse) moduli scheme for this type of abelian varieties.

1.13

Let G^i ($i = 1, 2$) be two reductive groups, $h^i : \mathbb{S} \rightarrow G_{\mathbb{R}}^i$ homomorphisms satisfying the conditions of 1.5 and X^i the set of conjugates of h^i by an element of $G^i(\mathbb{R})$. Let $G = G^1 \times G^2$, $h : \mathbb{S} \rightarrow G_{\mathbb{R}}$ the morphism (h^1, h^2) and $X = X^1 \times X^2$ the set of conjugates of h . In an obvious way we define an isomorphism

$$(1.13.1) \quad M_{\mathbb{C}}(G^1, h^1) \times M_{\mathbb{C}}(G^2, h^2) = M_{\mathbb{C}}(G, h).$$

For K^i compact open in $G^i(\mathbb{A}^f)$ and $K = K^1 \times K^2$, we similarly have

$$(1.13.2) \quad {}_{K^1} M_{\mathbb{C}}(G^1, h^1) \times {}_{K^2} M_{\mathbb{C}}(G^2, h^2) = {}_K M_{\mathbb{C}}(G, h).$$

1.14

Let G^1, G^2, h^1, h^2 be as in 1.13. We denote by $u : (G^1, h^1) \rightarrow (G^2, h^2)$ a homomorphism $u : G^1 \rightarrow G^2$ such that $uX^1 \subset X^2$. Such a homomorphism defines

$$(1.14.1) \quad u : M_{\mathbb{C}}(G^1, h^1) \rightarrow M_{\mathbb{C}}(G^2, h^2).$$

For K^i compact open in $G^i(\mathbb{A}^f)$, with $u(K^1) \subset K^2$, it defines

$$(1.14.2) \quad u(K^1, K^2) : {}_{K^1} M_{\mathbb{C}}(G^1, h^1) \rightarrow {}_{K^2} M_{\mathbb{C}}(G^2, h^2).$$

Proposition. 1.15 With the notations of 1.14, suppose that G^1 is a subgroup of G^2 . Then, for every compact open subgroup K^1 of $G^1(\mathbb{A}^f)$, there exists a compact open subgroup $K^2 \supset K^1$ of $G^2(\mathbb{A}^f)$ such that $u(K^1, K^2)$ identifies ${}_{K^1} M_{\mathbb{C}}(G^1, h^1)$ with a closed subscheme of ${}_{K^2} M_{\mathbb{C}}(G^2, h^2)$.

If K^i is small enough so that the map from $X^i \times (K^i \backslash G^i(\mathbb{A}^f))$ to ${}_{K^i} M_{\mathbb{C}}(G^i, h^i)$ is étale, then the map $u(K^1, K^2)$ is finite and unramified. It suffices to treat this case, and to show that $u(K^1, K^2)$ is injective for $K^2 \supset K^1$ sufficiently small. The graph of the equivalence relation $u(K^1, K^2)(x) = u(K^1, K^2)(y)$ is a closed subscheme of $({}_{K^1} M_{\mathbb{C}}(G^1, h^1))^2$, which decreases as K^2 shrinks, hence becomes stationary for $K^2 \supset K^1$ small enough. It suffices to show that

$$u(K^1) : {}_{K^1} M_{\mathbb{C}}(G^1, h^1) \rightarrow \varprojlim_{K^2 \supset K^1} {}_{K^2} M_{\mathbb{C}}(G^2, h^2) =_{\text{def}} {}_{K^1} M_{\mathbb{C}}(G^2, h^2)$$

is injective. We have

$$\begin{aligned} {}_{K^1} M_{\mathbb{C}}(G^1, h^1) &= K^1 \backslash M_{\mathbb{C}}(G^1, h^1) \\ {}_{K^1} M_{\mathbb{C}}(G^2, h^2) &= K^1 \backslash M_{\mathbb{C}}(G^2, h^2), \end{aligned}$$

so that it suffices to prove the variant:

Variant 1.15.1.

$$u : M_{\mathbb{C}}(G^1, h^1) \rightarrow M_{\mathbb{C}}(G^2, h^2) \text{ is injective.}$$

This assertion follows easily from the two lemmas below.

Lemma 1.15.2. Let $U^i \subset C^i(\mathbb{Q})$ be the group of units of the centre of G^i , and let $G^i(\mathbb{Q})^\wedge = \varprojlim_n (G^i(\mathbb{Q})/(U^i)^n)$. Then

$$M_{\mathbb{C}}(G^i, h^i) = K_{\infty}^i \backslash G^i(\mathbb{A})/G^i(\mathbb{Q})^\wedge.$$

This is a corollary of Chevalley's theorem that the $(U^i)^n$ are congruence subgroups of U^i .

Lemma 1.15.3. The map

$$u : G^1(\mathbb{A}^f)/G^1(\mathbb{Q})^\wedge \rightarrow G^2(\mathbb{A}^f)/G^2(\mathbb{Q})^\wedge$$

is injective.

By left translation, it suffices to show that $u^{-1}(\{e\}) = \{e\}$: one contemplates the diagram

$$\begin{array}{ccccccc} 0 & \longrightarrow & G^1(\mathbb{Q})^\wedge & \longrightarrow & G^2(\mathbb{Q})^\wedge & \longrightarrow & G^2/G^1(\mathbb{Q})^\wedge \\ & & \downarrow & & \downarrow & & \downarrow \\ 0 & \longrightarrow & G^1(\mathbb{A}^f) & \longrightarrow & G^2(\mathbb{A}^f) & \longrightarrow & G^2/G^1(\mathbb{A}^f) \\ & & \downarrow & & \downarrow & & \\ & & G^1(\mathbb{A}^f)/G^1(\mathbb{Q})^\wedge & \longrightarrow & G^2(\mathbb{A}^f)/G^2(\mathbb{Q})^\wedge & & \end{array}$$

where

$$G^2/G^1(\mathbb{Q})^\wedge = \varprojlim_n (G^2/G^1(\mathbb{Q})/(U^2/U^1)^n).$$

Chapter 2

Connected components

2.1

Let G be a connected reductive group over $\mathbb{Q}$. We retain the notations of 1.1 and assume that G' has no non-trivial normal subgroup H (defined over $\mathbb{Q}$) such that $H(\mathbb{R})$ is compact. This hypothesis does not exclude the possibility that $G'_{\mathbb{R}}$ has compact factors. Denote by $\widetilde{G}'$ the simply connected covering of the derived group G' .

Proposition. 2.2

- (i) The group $G(\mathbb{A}^f)$ acts transitively on the set $\pi_0(G(\mathbb{A})/G(\mathbb{Q}))$ of connected components of $G(\mathbb{A})/G(\mathbb{Q})$.
- (ii) The group $\widetilde{G}'(\mathbb{A})$ acts trivially on $\pi_0(G(\mathbb{A})/G(\mathbb{Q}))$.
- (iii) The quotient of $G(\mathbb{A})$ by the image of $\widetilde{G}'(\mathbb{A})$ is a commutative group.

Assertion (i) follows from (0.4): $G(\mathbb{A}) = G(\mathbb{A}^f)G(\mathbb{R})^0G(\mathbb{Q})$.

Since $\widetilde{G}$ is simply connected, the group $\widetilde{G}(\mathbb{R})$ is connected. By (0.3), $\widetilde{G}(\mathbb{R}) \cdot \widetilde{G}(\mathbb{Q})$ is dense in $\widetilde{G}(\mathbb{A})$, hence $\widetilde{G}(\mathbb{A})/\widetilde{G}(\mathbb{Q})$ is connected. For $x \in G(\mathbb{A})$, we have $\widetilde{G}(\mathbb{A}) \cdot x = x \cdot \widetilde{G}(\mathbb{A})$, so the image of $\widetilde{G}(\mathbb{A}) \cdot x$ in $G(\mathbb{A})/G(\mathbb{Q})$ is an image of $\widetilde{G}(\mathbb{A})/\widetilde{G}(\mathbb{Q})$: it is connected, and (ii) follows.

Let Z be the centre of $\widetilde{G}'$. Assertion (iii) follows from the fact that the “commutator” map $(x, y) \mapsto xyx^{-1}y^{-1} : G \times G \rightarrow G$ factors as

$$G \times G \longrightarrow G/C \times G/C \xleftarrow{\sim} \widetilde{G}'/Z \times \widetilde{G}'/Z \longrightarrow \widetilde{G}' \longrightarrow G.$$

Definition. 2.3 We denote by $\pi(G)$ the (commutative) quotient of $G(\mathbb{A})$ (or $G(\mathbb{A}^f)$) which acts faithfully on $\pi_0(G(\mathbb{A})/G(\mathbb{Q}))$.

In other words, $\pi(G)$ is $\pi_0(G(\mathbb{A})/G(\mathbb{Q}))$, equipped with its “obvious” commutative group structure. Recall the following theorem:

Theorem. 2.4 Under the hypotheses of 2.1, if G' is simply connected, then the canonical homomorphism

$$(2.4.1) \quad \pi(G) = \pi_0(G(\mathbb{A})/G(\mathbb{Q})) \longrightarrow \pi_0(T(\mathbb{A})/T(\mathbb{Q}))$$

is bijective.

To prove the surjectivity of (2.4.1), it suffices, by 2.2(i) applied to T , to show that $G(\mathbb{A}^f)$ maps onto $T(\mathbb{A}^f)$. The kernel G' of $\nu : G \rightarrow T$ being connected, we know that for almost all primes p ,

$\nu(G(\mathbb{Z}_p)) = T(\mathbb{Z}_p)$ (this formula makes sense for almost all p and is a corollary of Lang's theorem applied to G' reduced modulo p). It therefore suffices to prove that for every p , $\nu(G(\mathbb{Q}_p)) = T(\mathbb{Q}_p)$, which follows from (0.2) applied to G' .

Theorem 2.4 is equivalent to the more concrete statement:

Variant 2.5. For every compact open subgroup K of $G(\mathbb{A}^f)$, the map

$$(2.5.1) \quad \nu : \pi_0(K \backslash G(\mathbb{A})/G(\mathbb{Q})) \longrightarrow \pi_0(\nu(K) \backslash T(\mathbb{A})/T(\mathbb{Q}))$$

is bijective.

Indeed, $\pi_0(G(\mathbb{A})/G(\mathbb{Q})) = \varprojlim \pi_0(K \backslash G(\mathbb{A})/G(\mathbb{Q}))$, and similarly for T .

Moreover, $\pi_0(K \backslash G(\mathbb{A})/G(\mathbb{Q})) = G(\mathbb{R})^0 \times K \backslash G(\mathbb{A})/G(\mathbb{Q})$. We show that if $x, y \in G(\mathbb{A})$ have the same image in the right-hand side of (2.5.1), then they have the same image in the left-hand side. Left translation by y^{-1} , which replaces K by $y^{-1}Ky$, allows us to assume $y = e$. Modifying x by an element of $G(\mathbb{R})^0 \times K$, we may assume that $\nu(x) = \tau$ with $\tau \in T(\mathbb{Q})$. By the Hasse principle (0.1) for G' , the equation $\nu(\gamma) = \tau$ has a solution in $G(\mathbb{Q})$. Correcting x by γ , we may assume $x \in G'(\mathbb{A})$. By 2.2(ii), the image of x in $G(\mathbb{A})/G(\mathbb{Q})$ lies in the neutral component, and a fortiori x and y have the same image in the left-hand side of 2.5.1.

This proof seems to use (0.1), i.e. that G' has no E_8 factor. In fact, the class in $H^1(\mathbb{Q}, G')$ that obstructs the equation $\nu(\gamma) = \tau$ comes from the centre of G' , and the E_8 factors, being adjoint, count for nothing.

Proposition. 2.6 We have $K_\infty = C(\mathbb{R}) \cdot (K_\infty \cap G'(\mathbb{R})^0)$. The group $K_\infty \cap G'(\mathbb{R})^0$ is connected, and

$$\pi_0(K_\infty) \xrightarrow{\sim} \text{Im}(\pi_0(C(\mathbb{R}))) \longrightarrow \pi_0(G(\mathbb{R})).$$

Let $h' : \mathbb{S} \rightarrow G_{\mathbb{R}} \rightarrow (G/C)_{\mathbb{R}}$. The centralizer of h' in $(G/C)(\mathbb{R})$ is connected, because it is a compact group whose complexification is connected (as the centralizer of a torus). In particular, the image of K_∞ in $(G/C)(\mathbb{R})$ falls into the neutral component, which is the image of $G'(\mathbb{R})^0$, and $K_\infty = C(\mathbb{R})(K_\infty \cap G'(\mathbb{R})^0)$. The group $K_\infty \cap G'(\mathbb{R})^0$ is connected as a maximal compact subgroup of a connected group. Thus 2.6 follows.

2.7

The image of $\pi_0(K_\infty)$ in $\pi_0(G(\mathbb{A})/G(\mathbb{Q}))$ depends only on the conjugacy class of h , and $\pi_0(M_{\mathbb{C}}(G, h)) = \varprojlim \pi_0(K_\infty \times K \backslash G(\mathbb{A})/G(\mathbb{Q}))$ is identified with $\pi(G)/\pi_0(K_\infty)$.

In particular, if G' is simply connected, we have

$$(2.7.1) \quad \pi_0(M_{\mathbb{C}}(G, h)) \xrightarrow{\sim} \pi_0(T(\mathbb{A})/T(\mathbb{Q}))/\pi_0(K_\infty).$$

More concretely, for any compact open subgroup K of $G(\mathbb{A}^f)$, $\pi_0(K M_{\mathbb{C}}(G, h))$ is then a principal homogeneous space under $\nu(K \times K_\infty) \backslash T(\mathbb{A})/T(\mathbb{Q})$.

Chapter 3

Models

Let G be a reductive group over $\mathbb{Q}$, $h : \mathbb{S} \rightarrow G_{\mathbb{R}}$ satisfying the conditions of 1.5, and E a field equipped with a complex embedding $\rho : E \rightarrow \mathbb{C}$. Set $M_{\mathbb{C}}(G) = M_{\mathbb{C}}(G, h)$.

Definition 3.1

A model over E of $M_{\mathbb{C}}(G)$ consists of

- (a) a scheme M over E , equipped with a continuous action of $G(\mathbb{A}^f)$;
- (b) an isomorphism $m : M \otimes_{E, \rho} \mathbb{C} \rightarrow M_{\mathbb{C}}(G)$ compatible with the action of $G(\mathbb{A}^f)$.

Let F be a finite extension of E , equipped with a complex embedding extending that of E . If $M_E(G, h)$ is a model of $M_{\mathbb{C}}(G, h)$ over E , we denote by $M_F(G, h)$ the model $M_E(G, h) \otimes_E F$ of $M_{\mathbb{C}}(G, h)$ over F .

Remark 3.2 Giving a scheme M over E equipped with a continuous action of $G(\mathbb{A}^f)$ amounts to giving:

- (a) for every compact open subgroup K of $G(\mathbb{A}^f)$, a scheme ${}_K M$ over E ;
- (b) for two compact open subgroups K, L of $G(\mathbb{A}^f)$ and for $x \in G(\mathbb{A}^f)$ such that $xKx^{-1} \subset L$, a homomorphism $J_{L, K}(x) : {}_K M \rightarrow {}_L M$, these homomorphisms satisfying:
 - (α) $J_{M, L}(y) \circ J_{L, K}(x) = J_{M, K}(yx)$;
 - (β) $J_{K, K}(x) = \text{Id}$ if $x \in K$;
 - (γ) for K normal in L , J defines an action of L/K on ${}_K M$, and $J_{L, K}(e)$ gives $(L/K) \backslash {}_K M \xrightarrow{\sim} {}_L M$.

We will have

$$(3.2.1) \quad M = \varprojlim {}_K M \quad \text{and}$$

$$(3.2.2) \quad {}_K M = K \backslash M.$$

3.3

Let $M(G)$ be a model over E of $M_{\mathbb{C}}(G)$. For K compact open in $G(\mathbb{A}^f)$, set ${}_KM(G) = K \backslash M(G)$. The ${}_KM(G)$ are quasi-projective schemes over E , not necessarily geometrically connected. The structural isomorphism m induces isomorphisms

$$(3.3.1) \quad {}_KM(G) \otimes_{E,\rho} \mathbb{C} \xrightarrow{\sim} {}_KM_{\mathbb{C}}(G).$$

3.4

Suppose that G satisfies the hypotheses of 2.1, and let (M, m) be a model of $M_{\mathbb{C}}(G, h)$ over E . Let $\overline{E}$ be the algebraic closure of E in $\mathbb{C}$. Since M is defined over E , the Galois group $\text{Gal}(\overline{E}/E)$ acts on the profinite set

$$\pi_0(M \otimes_E \overline{E}) = \varprojlim \pi_0({}_KM \otimes_E \overline{E}) \xrightarrow[m]{} \pi_0(M_{\mathbb{C}}(G, h)).$$

The group $G(\mathbb{A}^f)$ acts on $\pi_0(M \otimes_E \overline{E})$ via its quotient $\pi(G)/\pi_0(K_{\infty})$, and $\pi_0(M \otimes_E \overline{E})$ is a principal homogeneous space under the commutative group $\pi(G)/\pi_0(K_{\infty})$ (2.6). The action of $\text{Gal}(\overline{E}/E)$ commutes with that of $G(\mathbb{A}^f)$, hence with that of $\pi(G)/\pi_0(K_{\infty})$. The action of an element σ of $\text{Gal}(\overline{E}/E)$ can therefore only be translation by an element $\lambda(\sigma)$ of $\pi(G)/\pi_0(K_{\infty})$, and λ is a homomorphism

$$(3.4.1) \quad \lambda_M : \text{Gal}(\overline{E}/E) \rightarrow \pi(G)/\pi_0(K_{\infty}).$$

Suppose that E is a number field. Class field theory then identifies the maximal abelian quotient of $\text{Gal}(\overline{E}/E)$ with $\pi_0(E^*(\mathbb{A})/E^*(\mathbb{Q}))$, and (3.4.1) becomes

$$(3.4.2) \quad \lambda_M : \text{Gal}(\overline{E}/E)^{\text{ab}} = \pi_0(E^*(\mathbb{A})/E^*(\mathbb{Q})) \rightarrow \pi(G)/\pi_0(K_{\infty}).$$

In the special case where G' is simply connected, the homomorphism (3.4.2) is identified (2.7.1) with

$$(3.4.3) \quad \lambda_M : \pi_0(E^*(\mathbb{A})/E^*(\mathbb{Q})) \rightarrow \pi_0(T(\mathbb{A})/T(\mathbb{Q}))/\pi_0(K_{\infty}).$$

Definition 3.5 The homomorphisms (3.4.1), (3.4.2), (3.4.3) are called the reciprocity law of the model M . If G' is simply connected, and λ_M comes from a homomorphism of algebraic groups over $\mathbb{Q}$ from E^* to T , this homomorphism is also called the "reciprocity law" of M .

3.6

There is a rule, applicable to each of the models constructed by Shimura, to determine E and the reciprocity law λ_M from G and h . The field E , for arbitrary G , and λ_M , for G' simply connected, are described as follows.

3.7

The composite homomorphism

$$hr : \mathbb{G}_m \xrightarrow{r} \mathbb{S}_{\mathbb{C}} \xrightarrow{h} G_{\mathbb{C}}$$

is a homomorphism over $\mathbb{C}$ between algebraic groups defined over $\mathbb{Q}$. The subfield E of $\mathbb{C}$ is the field of definition $E(G, h)$ of the conjugacy class of this homomorphism.

For G semi-simple adjoint, the field $E(G, h)$ can be described as follows. Let Δ be the Dynkin diagram of $G_{\mathbb{C}}$ and $|\Delta|$ be the set of vertices of Δ . The Galois group $\text{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$ acts on Δ .

Let H be a maximal torus of $G_{\mathbb{C}}$, equipped with a system of simple roots $(\alpha_i)_{i \in |\Delta|}$. The α_i identify H with $\mathbb{G}_{m, \mathbb{C}}^{|\Delta|}$. For any homomorphism $u : \mathbb{G}_{m, \mathbb{C}} \rightarrow G_{\mathbb{C}}$, there then exists a unique homomorphism $u' : \mathbb{G}_{m, \mathbb{C}} \rightarrow H = G_{m, \mathbb{C}}^{|\Delta|}$: $x \mapsto (x^{n_i})_{i \in |\Delta|}$ with $n_i \geq 0$, which is conjugate to u . The construction $u \mapsto \underline{n}(u) = (n_i)_{i \in |\Delta|}$ identifies the set of conjugacy classes of homomorphisms from $\mathbb{G}_{m, \mathbb{C}}$ to $G_{\mathbb{C}}$ with the set of functions from $|\Delta|$ to $\mathbb{N}$. It follows that the field $E(G, h)$ is defined by the subgroup of $\text{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$ that stabilises $\underline{n}(hr)$.

Proposition. 3.8 Let $G_{\mathbb{C}} = \prod_{i=1}^g G_i$ be the decomposition of the adjoint group G/C into $\mathbb{Q}$ -simple factors, and let h_i be the composite $h_i : \mathbb{S} \rightarrow G_{\mathbb{R}} \rightarrow G_{i, \mathbb{R}}$.

- (i) The subfield $E(G, h)$ of $\mathbb{C}$ is the compositum of the subfields $E(T, \nu \circ h)$ and $E(G_i, h_i)$ ($1 \leq i \leq g$) of $\mathbb{C}$.
- (ii) If the opposition involution of the Dynkin diagram of G_i respects $\underline{n}(h_i r)$, then $E(G_i, h_i)$ is totally real. Otherwise, $E(G_i, h_i)$ is a totally imaginary quadratic extension of a totally real number field.

Assertion (i) is easy to verify, and it suffices to prove (ii) for G $\mathbb{Q}$ -simple and adjoint. Let Δ be its Dynkin diagram. It follows from hypotheses 1.5 that $G_{\mathbb{R}}$ admits a maximal compact torus; this implies that complex conjugation acts on Δ by the opposition involution ι . This involution lies in the centre of the automorphism group of Δ .

Let E' be the Galois extension of $\mathbb{Q}$ defined by the subgroup of $\text{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$ that acts trivially on Δ . Then, the image σ of the complex conjugation is central in $\text{Gal}(E'/\mathbb{Q})$; It follows that E' is either totally real or a totally imaginary quadratic extension of a totally real field. Finally, σ is the identity on $E(G, h) \subset E'$ if and only if ι respects $\underline{n}(hr)$, whence the assertion.

3.9

We retain the notations of 3.6. Assume that G' is simply connected, that E contains $E(G, h)$, and that the complex embedding of E induces that of $E(G, h)$. The composite morphism

$$r''(h) = \nu hr : \mathbb{G}_m \xrightarrow{r} \mathbb{S}_{\mathbb{C}} \xrightarrow{h} G_{\mathbb{C}} \xrightarrow{\nu} T_{\mathbb{C}}$$

depends only on the conjugacy class of hr . It is therefore defined over E , i.e. comes by base change from E to $\mathbb{C}$ from a homomorphism of algebraic groups over E , still denoted $r''(h) : \mathbb{G}_m \rightarrow T_E$. Applying Weil restriction of scalars, and let $r'(h) : E^* \rightarrow T$ be the composite

$$E^* = R_{E/\mathbb{Q}}(\mathbb{G}_{m, E}) \xrightarrow{R_{E/\mathbb{Q}}(r''(h))} R_{E/\mathbb{Q}}(T_E) \xrightarrow{N_{E/\mathbb{Q}}} T.$$

The reciprocity law of M is the inverse $r(G, h)$ of $r'(h)$:

$$(3.9.1) \quad r(G, h) = r'(h)^{-1} : E^* \rightarrow T.$$

3.10

Example II. Let H be a torus and $h : \mathbb{S} \rightarrow H_{\mathbb{R}}$. The conditions of 1.5 are automatically satisfied. The spaces $M_{\mathbb{C}}(H, h)$ are finite sets. A reduced finite scheme over the field $E(H, h)$ (3.7) corresponds to a Galois set. It follows that there exists, up to a unique isomorphism, one and only one model of $M_{\mathbb{C}}(H, h)$ over $E(H, h)$, with reciprocity law given by 3.9.1. We denote it by $M(H, h)$.

3.11

Let F be a finite extension of E , equipped with a complex embedding extending that of E . The diagram

$$\begin{array}{ccc} F^* & \xrightarrow{N_{F/E}} & E^* \\ & \searrow r(G, h) & \swarrow r(G, h) \\ & T & \end{array}$$

is commutative. If $M(G, h)$ is a model over E of $M_{\mathbb{C}}(G, h)$ having (3.9.1) as reciprocity law, it follows that $M_F(G, h)$ (3.1) also has (3.9.1) as reciprocity law.

3.12

Let $u : (G^1, h^1) \rightarrow (G^2, h^2)$ be as in 1.14, and let $M_{E^i}(G^i, h^i)$ be a model of $M_{\mathbb{C}}(G^i, h^i)$ over E^i . Let E be the compositum of the subfields E^1 and E^2 of $\mathbb{C}$. We have (notation of 3.1)

$$M_E(G^i, h^i) \otimes_E \mathbb{C} = M_{\mathbb{C}}(G^i, h^i) \quad (i = 1, 2).$$

It therefore makes sense to ask whether

$$u : M_{\mathbb{C}}(G^1, h^1) \rightarrow M_{\mathbb{C}}(G^2, h^2)$$

is defined over E .

Definition 3.13 Let G be a connected reductive group over $\mathbb{Q}$ satisfying the condition of 2.1, $h : \mathbb{S} \rightarrow G_{\mathbb{R}}$ a homomorphism satisfying the conditions of 1.5, and E an extension of $E(G, h)$ (3.7) equipped with a complex embedding extending that of $E(G, h)$. A model $M(G, h)$ of $M_{\mathbb{C}}(G, h)$ over E is called weakly canonical if, for every torus $u : H \hookrightarrow G$ equipped with $h' : \mathbb{S} \rightarrow H_{\mathbb{R}}$ such that uh' is conjugate to h by an element of $G(\mathbb{R})$, the morphism (1.14)

$$M_{\mathbb{C}}(H, h') \rightarrow M_{\mathbb{C}}(G, h)$$

is defined (3.10) (3.12) over the compositum of $E(H, h')$ and E inside $\mathbb{C}$. A model $M_E(G, h)$ is canonical if it is weakly canonical and that $E = E(G, h)$.

3.14

Translation. Let M be a canonical model of $M_{\mathbb{C}}(G, h)$, with reciprocity law λ_M . If G' is simply connected, then λ_M is given by 3.9.1 (5.6). For every compact open subgroup K of $G(\mathbb{A}^f)$, denote

again by $\nu(K)$ the image of K in $\pi(G)/\pi_0(K_\infty)$, and let $E(K)$ be the extension of $E(G, h)$ defined by the idèle class group $\lambda_M^{-1}(\nu(K))$. One checks that $E(K)$ is the field of definition of the neutral component ${}_KM_{\mathbb{C}}^0$ of ${}_KM_{\mathbb{C}}$, considered as a subscheme of ${}_KM_{\mathbb{C}} = {}_KM \otimes_{E, \rho} \mathbb{C}$.

By definition, ${}_KM_{\mathbb{C}}^0$ comes by base change from $E(K)$ to $\mathbb{C}$ from a subscheme ${}_KM'$ of ${}_KM \otimes_E E(K)$. The scheme ${}_KM'/E(K)$ is geometrically connected over $E(K)$, and one can also describe the scheme ${}_KM'$ as that of the connected components (over E) of ${}_KM$ which, after extension of scalars from E to $\mathbb{C}$, contains the origin.

$$\begin{array}{ccccc}
 {}_KM & \longleftrightarrow & {}_KM' & \longleftarrow & {}_KM_{\mathbb{C}}^0 \\
 \downarrow & & \downarrow & & \downarrow \\
 & & \text{Spec}(E(K)) & \longleftarrow & \text{Spec}(\mathbb{C}) \\
 \downarrow & & \downarrow & & \\
 \text{Spec}(E) & \xlongequal{\quad} & \text{Spec}(E) & &
 \end{array}$$

Shimura has the habit to express himself in terms of the system of varieties ${}_KM'/E(K)$ and the homomorphisms of type 2.2 between them. For a typical statement, see [14, p.146]. For the translation, cf. [14, 2.7].

3.15

For $u : (H, h') \rightarrow (G, h)$ as in 3.13, the image of $M_{\mathbb{C}}(H, h')$ in ${}_KM_{\mathbb{C}}(G, h)$ is a finite set. The hypothesis is that this subset of ${}_KM_{\mathbb{C}}(G, h)$ is defined over $E(H, h')$, that its points are defined over an abelian extension of $E(H, h')$, and that $\text{Gal}(\overline{E}(H, h')/E(H, h'))$ permutes them in a prescribed manner.

We will call "special" the points of ${}_KM(G, h)$ that lie in the image of some ${}_KM(H, h')$.

Suppose we know a projective embedding $q : {}_KM_{\mathbb{C}}^0 \rightarrow \mathbb{P}^r$ that is defined over $E(K)$. This can be interpreted as $\tilde{q} : X^0 \rightarrow \mathbb{P}^r$ (cf. 1.7, 1.8). If $uh' \in X$ lies in X^0 , then the coordinates of $\tilde{q}(uh')$ generate over $E(K)E(H, h')$ an abelian extension of $E(H, h')$ which can be made explicit as a class field.

3.16

The classical example of such a situation is provided by the moduli scheme of elliptic curves (corresponding to $G = \text{GL}_2$, $K = \text{GL}_2(\widehat{\mathbb{Z}})$, h as in 1.6), and by those points defined by elliptic curves with complex multiplication.

Let j be the modular invariant, viewed as a function on the Poincaré upper half-plane X^0 . If $\tau \in X^0$ belongs to an imaginary quadratic field K , let $L(\tau) = \mathbb{Z} \oplus \mathbb{Z}\tau \subset \mathbb{C}$, and let $\mathcal{O}(\tau) = \{k \in K \mid kL(\tau) \subset L(\tau)\}$. The field $K(j(\tau))$ is the abelian extension of K with Galois group the class group of ideals of $\mathcal{O}(\tau)$. If σ in Galois corresponds to an invertible module L , and if $L(\tau') \simeq L(\tau) \otimes_{\mathcal{O}(\tau)} L$, then $j(\tau) = j(\tau')^\sigma$.

For more explicit, stranger examples, see [15, pp. 45-46].

Chapter 4

Abelian varieties

4.1

Let k be an algebraically closed field of characteristic 0. Let A be an abelian variety over k . For $n \in \mathbb{N}^+$, denote by A_n the kernel of multiplication by n . The A_n form a projective system ($\varphi_{nm,n} : A_{nm} \rightarrow A_n$ is $x \mapsto x^m$) and we set

$$\hat{T}(A) = \varprojlim A_n, \quad \hat{V}(A) = \mathbb{A}^f \otimes_{\hat{\mathbb{Z}}} \hat{T}(A).$$

The $\hat{\mathbb{Z}}$ -module $\hat{T}(A)$ is the product of the Tate modules $T_\ell(A)$. If $k = \mathbb{C}$, we have

$$(4.1.1) \quad \hat{T}(A) = \hat{\mathbb{Z}} \otimes H_1(A, \mathbb{Z}),$$

$$(4.1.2) \quad \hat{V}(A) = \mathbb{A}^f \otimes H_1(A, \mathbb{Q}).$$

4.2

The category of abelian varieties up to isogeny over k is the category whose objects are abelian varieties over k and in which $\text{Hom}_{\text{iso}}(A, B) = \text{Hom}(A, B) \otimes \mathbb{Q}$. We denote by $A \otimes \mathbb{Q}$ the abelian variety up to isogeny "underlying" an abelian variety A . Any additive functor from the category of abelian varieties to a $\mathbb{Q}$ -linear additive category factors through the functor $A \mapsto A \otimes \mathbb{Q}$. Thus, the contravariant functor "dual variety" $A \mapsto A^*$ extends to abelian varieties up to isogeny. The functor $A \mapsto \hat{V}(A)$ extends similarly. Moreover, if A_0 is an abelian variety up to isogeny, giving a B together with an isomorphism $B \otimes \mathbb{Q} = A_0$ is equivalent to giving $\hat{T}(B) \subset \hat{V}(A_0)$.

4.3

Let A be an abelian variety over k . Denote by $\text{NS}(A)$ the group of algebraic equivalence classes of invertible sheaves on A . We define an effective polarization (resp. a polarization) of A as an element of $\text{NS}(A)$ (resp. of $\text{NS}(A) \otimes \mathbb{Q}$) a positive multiple of which is defined by a projective embedding of A . A homogeneous polarization is an element of $(\text{NS}(A) \otimes \mathbb{Q})/\mathbb{Q}^*$ that is the class of a polarization.

Recall that $\text{NS}(A) \otimes \mathbb{Q}$ is identified via $p \mapsto p'$ with the set of elements of $\text{Hom}(A, A^*) \otimes \mathbb{Q}$ equal to their own transpose. An isomorphism $u \in \text{Hom}(A \otimes \mathbb{Q}, B \otimes \mathbb{Q})$ induces isomorphisms from

$\text{Hom}(A, A^*) \otimes \mathbb{Q}$ to $\text{Hom}(B, B^*) \otimes \mathbb{Q}$, from $\text{NS}(A) \otimes \mathbb{Q}$ to $\text{NS}(B) \otimes \mathbb{Q}$, and sends polarizations to polarizations. This allows us to speak of polarizations of an abelian variety up to isogeny.

4.4

A polarization p of A defines in $\text{End}(A) \otimes \mathbb{Q}$ a positive involution $u \mapsto p'^{-1}u^*p'$, which depends only on the homogeneous polarization $\mathbb{Q}^* \cdot p$.

Let F be a product of totally real fields, and let $\rho : F \rightarrow \text{End}(A) \otimes \mathbb{Q}$. Let $\text{NS}(A)_\rho \subset \text{NS}(A)$ be the set of p such that $p' \cdot \rho(f) = \rho(f)^* \cdot p'$ for all $f \in F$. The vector space $\text{NS}_\rho(A) \otimes \mathbb{Q}$ is equipped with, for $f \in F$, an action of F : $(f \cdot p)' = p' \circ \rho(f) = \rho(f)^* \circ p'$. A weak polarization of A (relative to ρ, F) is an element of $(\text{NS}_\rho(A) \otimes \mathbb{Q})/F^*$ that is the class of a polarization. The set of weak polarizations of A depends only on $A \otimes \mathbb{Q}$. The restriction to the centralizer of F of the involution of $\text{End}(A) \otimes \mathbb{Q}$ defined by a polarization $p \in \text{NS}_\rho(A)$ depends only on the weak polarization $F^* \cdot p$.

4.5

If A_0^* is the dual of an abelian variety up to isogeny A_0 , then $\hat{V}(A_0)$ and $\hat{V}(A_0^*)$ are in duality with values in $\mathbb{A}^f(1)$. If p is a polarization of A_0 , the polarization form $\psi_p(x, y) = \langle x, p'(y) \rangle$ is a non-degenerate alternating form on the free $\mathbb{A}^f$ -module $\hat{V}(A_0)$, with values in $\mathbb{A}^f(1)$.

The reminders 4.1 and 4.5 extend to abelian schemes over an arbitrary base.

4.6

Take $k = \mathbb{C}$. The additive functor $A \mapsto H_1(A, \mathbb{Q})$ extends to abelian varieties up to isogeny over $\mathbb{C}$ (4.2). For A_0 an abelian variety up to isogeny, $H_1(A_0, \mathbb{Q})$ and $H_1(A_0^*, \mathbb{Q})$ are in duality. Via 4.1.2 and the isomorphism $\mathbb{A}^f(1) \simeq \mathbb{A}^f$ defined by the exponential map, this duality is compatible with that considered in 4.5. The alternating form $\psi_p(x, y) = \langle x, p'(y) \rangle$ on $H_1(A_0, \mathbb{Q})$ defined by a polarization p of A_0 is again called the polarization form. The Hodge structure of $H_1(A_0, \mathbb{Q})$ is the homomorphism $h : \mathbb{S} \rightarrow \text{GL}(H_1(A_0, \mathbb{Q}))_{\mathbb{R}}$ such that $\mathbb{S}$ acts on $H_1(A_0, \mathbb{Q}) \otimes \mathbb{C}$ by the characters z^{-1} and $\bar{z}^{-1}$, and that $F(h)^0$ (1.5) is the kernel of the exponential map $H_1(A_0, \mathbb{Q}) \otimes \mathbb{C} \rightarrow \text{Lie}(A)$.

Theorem 4.7 (Riemann).

The constructions

$$\begin{aligned} (A, p) &\mapsto (H_1(A, \mathbb{Q}), \psi_p, H_1(A, \mathbb{Z}), h), \\ (V, \psi, V_{\mathbb{Z}}, h) &\mapsto F(h)^0 \setminus V \otimes \mathbb{C}/V_{\mathbb{Z}} \end{aligned}$$

establish an equivalence between

- (a) polarized abelian varieties over $\mathbb{C}$;
- (b) systems consisting of a vector space V over $\mathbb{Q}$, a non-degenerate alternating form ψ on V , an integral lattice $V_{\mathbb{Z}}$ in V , and a homomorphism h from $\mathbb{S}$ to $\text{Gp}(V)_{\mathbb{R}}$ of symplectic similitudes of $V \otimes \mathbb{R}$ of the type in 1.6 such that

$$\psi(x, h(i)x) > 0 \quad (x \in V \otimes \mathbb{R}, x \neq 0).$$

Variant 4.8 Giving an abelian variety up to isogeny A over $\mathbb{C}$ equipped with a homogeneous polarization is equivalent to giving a vector space over $\mathbb{Q}$ equipped with a non-degenerate alternating form ψ up to a factor, and a homomorphism $h : \mathbb{S} \rightarrow \mathrm{Gp}(V)$ of type 1.6.

4.9

Let L be a semi-simple algebra with involution over $\mathbb{Q}$, and V a vector space over $\mathbb{Q}$ equipped with a faithful L -module structure and a non-degenerate alternating form ψ such that

$$(4.9.1) \quad \psi(\ell x, y) = \psi(x, \ell^* y).$$

Denote by G the algebraic group over $\mathbb{Q}$ of L -linear symplectic similitudes of V . The group $G(\mathbb{Q})$ is the set of $g \in \mathrm{GL}_L(V)$ such that there exists $\mu(g) \in \mathbb{Q}^*$ satisfying

$$(4.9.2) \quad \psi(gx, gy) = \mu(g)\psi(x, y).$$

We suppose given a homomorphism $h_0 : \mathbb{S} \rightarrow G_{\mathbb{R}}$ such that, via h_0 , $\mathbb{S}$ acts on $V_{\mathbb{C}}$ by the characters z^{-1} and $\bar{z}^{-1}$ and the form $\psi(x, h_0(i)y)$ is symmetric and positive definite. Conditions (1.5.1)-(1.5.3) are then satisfied by (G, h_0) , and the involution of L is positive.

4.10

Apply (1.11) to (G, h_0) and to the representation V of G . A G -structure $\bar{\iota}$ on a vector space H is interpreted, by 1.8, as the datum on H of an L -module structure and an alternating form ψ up to a factor, such that the vector space H with these structures is isomorphic to V . Let $\bar{\iota}$ be a G -structure on H . A datum 1.9 (b) $h : \mathbb{S} \rightarrow G_{\mathbb{R}}^{\bar{\iota}}$ on H then defines, by 4.8 applied to (H, ψ, h) , an abelian variety up to isogeny A equipped with a homogeneous polarization $\bar{p}$ such that $H = H_1(A, \mathbb{Q})$, h is the Hodge structure of $H_1(A, \mathbb{Q})$, and ψ is the polarization form. Moreover, the L -module structure on H comes from $\rho : L \rightarrow \mathrm{End}(A)$, and H equipped with $\bar{\iota}$ and h is determined by $(A, \bar{p}, \rho)$.

For a triple $(A, \bar{p}, \rho)$ to come from an H as above, it is necessary and sufficient that

$$(4.10.1) \quad H_1(A, \mathbb{Q}) \text{ with its } L\text{-module structure and polarization form (up to a factor) is isomorphic to } V;$$

$$(4.10.2) \quad \text{For an isomorphism } \iota : V \rightarrow H_1(A, \mathbb{Q}), \iota^{-1}h\iota \text{ is conjugate to } h_0.$$

Let $t : L \rightarrow \mathbb{C}$ be the map given by

$$(4.10.3) \quad t(\ell) = \mathrm{Tr}(\ell; V_{\mathbb{C}}/F^0(h_0)).$$

Annotation 4.11 Let K be a compact open subgroup of $G(\mathbb{A}^f)$. The points of ${}_K M_{\mathbb{C}}(G, h_0)$ correspond bijectively to the isomorphism classes of abelian varieties up to isogeny A , equipped with

(a) $\rho : L \rightarrow \mathrm{End}(A)$ such that

$$\mathrm{Tr}(\rho(\ell), \mathrm{Lie}(A)) = t(\ell) \quad (\ell \in L),$$

(b) a homogeneous polarization $\bar{p}$ that induces the given involution of L ,

(c) a class modulo $K, \bar{k}$, of L -linear symplectic similitudes $k : \hat{V}(A) \xrightarrow{\sim} V \otimes \mathbb{A}^f$,

and such that

(*) the conditions (4.10.1) and (4.10.2) are satisfied.

We apply 1.11 and 4.10, noting that a datum 1.9(c) is identified with a datum (c), and that the conditions in (b) and (a) follow from (4.10.1) and (4.10.2) respectively.

4.12

Let K be a compact open subgroup of $G(\mathbb{A}^f)$, $V_{\mathbb{Z}}$ an integral lattice in V such that $V_{\mathbb{Z}} = V_{\mathbb{Z}} \otimes \hat{\mathbb{Z}}$ is K -invariant, $L_{\mathbb{Z}}$ an order of L such that $L_{\mathbb{Z}} V_{\mathbb{Z}} \subset V_{\mathbb{Z}}$, $\psi_{\mathbb{Z}}$ a positive rational multiple of the alternating form of V taking integer values on $V_{\mathbb{Z}}$, and $V'_{\mathbb{Z}}$ the largest lattice of V such that $\psi_{\mathbb{Z}}(V_{\mathbb{Z}}, V'_{\mathbb{Z}}) \subset \mathbb{Z}$. Let n be an integer, and $K_n = \{g \in G(\mathbb{A}^f) \mid (g-1)V_{\mathbb{Z}} \subset nV_{\mathbb{Z}}\}$. Assume n is large enough so that $K \supset K_n$.

If A is as in 4.11, then $k^{-1}(V_{\mathbb{Z}}) \subset \hat{V}(A)$ does not depend on the choice of $k \in \bar{k}$, and defines an abelian variety B with $B \otimes \mathbb{Q} \simeq A$ and $\hat{T}(B) = k^{-1}(V_{\mathbb{Z}}) \subset \hat{V}(A)$. The action of L on A induces an action of $L_{\mathbb{Z}}$ on B . There exists a unique effective polarization p of B , with $p \in \bar{p}$, such that $k^{-1}(V'_{\mathbb{Z}})$ is the largest “lattice” $\hat{T}'(B) \supset \hat{T}(B)$ satisfying $\psi_p(\hat{T}(B), \hat{T}'(B)) \subset \hat{\mathbb{Z}}(1)$. Finally, the set $\bar{k}$ of L -linear symplectic isomorphisms from $\hat{T}(B)$ to $V_{\mathbb{Z}}$ is the inverse image of its image $\bar{k}_n$ in $\text{Isom}(B_n, V_{\mathbb{Z}}/nV_{\mathbb{Z}})$.

This allows us to interpret the points of ${}_K M_{\mathbb{C}}(G, h_0)$ as corresponding to isomorphism classes of systems $(B, p, \rho, \bar{k}_n)$ consisting of

- (a) a polarized abelian variety (B, p) with complex multiplication by $L_{\mathbb{Z}}$, satisfying 4.11(a)(b);
- (b) a class modulo $K/K_n, \bar{k}_n$, of isomorphisms $k_n : B_n \xrightarrow{\sim} V_{\mathbb{Z}}/nV_{\mathbb{Z}}$ that can be lifted to L -linear symplectic isomorphisms $k : \hat{T}(B) \rightarrow V_{\mathbb{Z}}$, and satisfying (4.10.1) and (4.10.2).

4.13

Denote by F the algebra of invariants of the centre of L under $*$. It is a product of totally real fields. There exists one and only one F -bilinear alternating form ${}_F\psi$ on V such that $\text{Tr}_{F/\mathbb{Q}} \psi = \psi$.

Denote by G_1 the group of L -linear symplectic similitudes of ${}_F\psi$: $G_1(\mathbb{Q})$ is the set of $g \in \text{GL}_L(V)$ such that there exists $\mu(g) \in F^*$ satisfying

$$(4.13.1) \quad {}_F\psi(gx, gy) = \mu(g) {}_F\psi(x, y), \quad \text{i.e.}$$

$$(4.13.2) \quad \psi(gx, gy) = \psi(\mu(g)x, y).$$

As above, one verifies the

Variante 4.14 Let K be a compact open subgroup of $G_1(\mathbb{A}^f)$. The points of ${}_K M_{\mathbb{C}}(G_1, h_0)$ correspond bijectively to the isomorphism classes of abelian varieties up to isogeny A , equipped with

- (a) $\rho : L \rightarrow \text{End}(A)$ as in 4.11;
- (b) a weak polarization $\bar{p}$ (relative to F) that induces the given involution of L ;
- (c) a class modulo K of L -linear symplectic similitudes $k : \hat{V}(A) \xrightarrow{\sim} V \otimes \mathbb{A}^f$ (for the $F \otimes \mathbb{A}^f$ -bilinear forms on both sides), and such that
- (*) conditions analogous to (4.10.1)(4.10.2) are satisfied.

4.15

One can make 4.11 and 4.14 more precise by showing that ${}_KM_{\mathbb{C}}(G, h)$ is the coarse moduli scheme of an evident functor on schemes of finite type over $\mathbb{C}$.

4.16

Example I (continued from 1.6, 1.11). We place ourselves in the special case of 4.9 where $L = \mathbb{Q}$ and $\dim V = 2g$. Then $G = \mathrm{Gp}(V)$ (group of symplectic similitudes). For $h_0 : \mathbb{S} \rightarrow G_{\mathbb{R}}$ of type 1.6, we have $E(\mathrm{Gp}, h_0) = \mathbb{Q}$. Let $V_{\mathbb{Z}}, n$ and K_n be as in 1.11. For any scheme S , let $F(S)$ be the set of isomorphism classes of principally polarized abelian schemes A/S equipped with a symplectic similitude $k_n : A_n \xrightarrow{\sim} (V_{\mathbb{Z}}/nV_{\mathbb{Z}})_S$. For $n \geq 3$, the classified objects have no automorphisms and the functor F is represented by a $\mathbb{Q}$ -scheme ${}_K M$ [6]. By 1.11, 4.11 and 4.12, we have an isomorphism

$$m_n : {}_{K_n}M_{\mathbb{C}} \xrightarrow{\sim} {}_{K_n}M_{\mathbb{C}}(\mathrm{Gp}, h_0).$$

In the language of 3.1, we have

Proposition 4.17 The $\mathbb{Q}$ -scheme $M_{\mathbb{C}}(\mathrm{Gp}, h_0) = \varprojlim {}_{K_n}M_{\mathbb{C}}$ is a model over $\mathbb{Q} = E(\mathrm{Gp}, h_0)$ of $M_{\mathbb{C}}(\mathrm{Gp}, h_0)$.

4.18

We place ourselves in the special case of 4.9 where V is a cyclic L -module. Then the groups G and G_1 are contained in the centre of L^* .

Let $\bar{E}$ be an algebraic closure of $E(G, h)$ and let M^+ be the set of isomorphism classes $[A, \rho, k, \bar{p}]$ of objects $(A, \rho, k, \bar{p})$ consisting of

- (a) an abelian variety up to isogeny $A/\bar{E}$ equipped with a homogeneous polarization $\bar{p}$ and with $\rho : L \rightarrow \mathrm{End}(A)$ such that, with the notation of 4.10.3, we have for $\ell \in L$

$$\mathrm{Tr}(\ell, \mathrm{Lie}(A)) = t(\ell) \in E(G, h);$$

- (b) an L -linear isomorphism $k : \hat{V}(A) \xrightarrow{\sim} V \otimes \mathbb{A}^f$.

The abelian variety A is therefore "of CM type". We refer to [16] for the proof of the following fundamental theorem.

Theorem 4.19 (Shimura-Taniyama)

The Galois group $\mathrm{Gal}(\bar{E}/E(G, h))$ acts on M^+ through its maximal abelian quotient $\pi_0(E(G, h)^*(\mathbb{A})/E(G, h)^*(\mathbb{Q}))$.

For $\underline{e} \in E(G, h)^*(\mathbb{A})$, with image $\varphi(\underline{e})$ in the abelianized Galois group, and with component $\underline{e}^f$ in $E(G, h)^*(\mathbb{A}^f)$, we have

$$(4.19.1) \quad \varphi(\underline{e})([A, \rho, k, \bar{p}]) = [A, \rho, r(G, h)(\underline{e}^f) \cdot k, \bar{p}].$$

Let τ be an embedding of $\bar{E}$ into $\mathbb{C}$ extending that of $E(G, h)$, K a compact open subgroup of $G(\mathbb{A}^f)$, and ${}_KM(G, h)(\bar{E})$ the set of isomorphism classes of objects $(A, \rho, \bar{k}, \bar{p})$ where

- (a) A, ρ and $\bar{p}$ are as above;
- (b) $\bar{k}$ is a class modulo K of $L \otimes \mathbb{A}^f$ -linear symplectic similitudes $k : \hat{V}(A) \xrightarrow{\sim} V \otimes \mathbb{A}^f$;
- (c) the vector space $H_1(A_{\mathbb{C}}, \mathbb{Q})$ (where $A_{\mathbb{C}}$ is defined via τ), equipped with the action of L and the polarization form (given up to a factor), is isomorphic to V .

By 4.19, condition (c) is independent of the choice of τ , so that the finite set ${}_KM(G, h)(\bar{E})$ is equipped with an action of $\text{Gal}(\bar{E}/E(G, h))$; as a Galois set, it is identified with the set of points over $\bar{E}$ of a finite étale $E(G, h)$ -scheme ${}_KM(G, h)$. By 4.14, ${}_KM(G, h)(\mathbb{C}) = G(\mathbb{R}) \times K \backslash G(\mathbb{A})/G(\mathbb{Q})$, and by 4.19, we have:

Proposition 4.20 The $E(G, h)$ -scheme $M(G, h) = \varprojlim {}_KM(G, h)$ is a canonical model of $M_{\mathbb{C}}(G, h)$.

Theorem 4.21 The model $M(\text{Gp}, h_0)$ constructed in 4.17 is a canonical model.

Let (V, ψ) be as in 4.16, $L, \rho : L \rightarrow \text{End}(V)$, G and $h : \mathbb{S} \rightarrow G_{\mathbb{R}}$ as in 4.18. We have a morphism $u : (G, h) \hookrightarrow (\text{Gp}, h_0)$. Let $K \subset G(\mathbb{A}^f)$ such that $u(K) \subset K_n$ (4.16). If $\bar{E}$ is an algebraic closure of $E(G, h)$, we define

$$u : {}_KM(G, h)(\bar{E}) \rightarrow {}_{K_n}M(\text{Gp}, h_0)(\bar{E})$$

by "forgetting the complex multiplication", associating to $[A, \rho, \bar{k}, \bar{p}]$ the triple consisting of

- (a) the abelian variety B with $B \otimes \mathbb{Q} \simeq A$ and such that

$$\hat{T}(B) = k^{-1}(V_{\hat{\mathbb{Z}}}) \subset \hat{V}(A) \quad (k \in \bar{k});$$

- (b) the polarization $\bar{p}$ of B ;
- (c) the isomorphism $k : B_n \xrightarrow{\sim} V_{\mathbb{Z}}/nV_{\mathbb{Z}}$ ($k \in \bar{k}$).

The maps u define a morphism of models

$$u : M(G, h) \rightarrow M(\text{Gp}, h_0) \otimes E(G, h).$$

One completes the proof by showing ([8]) that for every $u : (H, h') \rightarrow (\text{Gp}, h_0)$ as in 3.13, there exists a $v : (G, h) \rightarrow (\text{Gp}, h_0)$ as above (for a suitable L) such that $uh' = vh$.

Variant 4.22 Let F be a totally real number field, $n = [F : \mathbb{Q}]$, and $\text{Gp}_{2g}(F)$ the group of symplectic similitudes of an F -vector space of dimension $2g$. Let $h_0 : \mathbb{S} \rightarrow \text{Gp}_{2g}(F)_{\mathbb{R}} \simeq \text{Gp}_{2g}(\mathbb{R})^n$ be as in 1.6. One can construct a canonical model of $M_{\mathbb{C}}(\text{Gp}_{2g}(F), h_0)$ from moduli of weakly polarized abelian schemes.

Chapter 5

Construction techniques

The definition 3.13 is usable only because there are "many" special points.

Theorem 5.1 Let G and h be as in 3.13. For every finite extension F of $E(G, h)$, there exists $(H, h', u : H \hookrightarrow G)$ as in 3.13 such that the extension $E(H, h')$ of $E(G, h)$ is linearly disjoint from F .

Let $Y = \underline{\text{Hom}}(\mathbb{G}_m, G)/G$ be the scheme of conjugacy classes of morphisms from $\mathbb{G}_m$ to G . The Galois group $\text{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$ acts on the (infinite discrete) set $Y(\mathbb{C})$ of conjugacy classes of morphisms from $\mathbb{G}_m$ to $G_{\mathbb{C}}$. Let Y_0 be the finite subscheme of Y such that $Y_0(\mathbb{C})$ is the Galois orbit of the class of hr . This is the spectrum of $E(G, h)$.

Let $V' = \text{Lie}(G)$, considered as an affine space over $\mathbb{Q}$, and let V be the open subscheme of regular elements of the Lie algebra. For $v \in V$, denote by T_v the centraliser of v (a maximal torus). Let W be the moduli scheme of maximal tori T of G , equipped with $s : \mathbb{G}_m \rightarrow T$ and $v \in \text{Lie}(T)$ such that

- (a) v is regular in $\text{Lie}(G)$;
- (b) the class of $s : \mathbb{G}_m \rightarrow G$ lies in Y_0 .

The morphism $f : W \rightarrow V : (T, s, v) \mapsto v$, is finite étale surjective (note that $T = T_v$). Let p be the morphism from W to $Y_0 : (T, s, v) \mapsto (\text{class of } s)$.

$$(5.1.1) \quad \begin{array}{ccc} W & \xrightarrow{\quad} & V \\ \downarrow p & & \downarrow \\ \text{Spec}(E(G, h)) & \longrightarrow & \text{Spec}(\mathbb{Q}) \end{array}$$

Lemma 5.1.2 The scheme W is irreducible, and the geometric fibres of p are (geometrically) irreducible.

Since Y_0 is irreducible (the spectrum of an extension of $\mathbb{Q}$), it suffices to prove the second assertion. It follows from:

- (a) because G is connected, the scheme over $\mathbb{C}$ of homomorphisms from $\mathbb{G}_{m, \mathbb{C}}$ to G conjugate to a given homomorphism is irreducible;

- (b) for a given $i : \mathbb{G}_{m,\mathbb{C}} \rightarrow G_{\mathbb{C}}$, by the conjugacy theorem for maximal tori applied to the (connected) centraliser of $i(\mathbb{G}_{m,\mathbb{C}})$, the scheme of maximal tori of $G_{m,\mathbb{C}}$ containing $i(\mathbb{G}_{m,\mathbb{C}})$ is irreducible.

Let $U \subset V(\mathbb{R})$ be the set of $v \in V(\mathbb{R})$ such that $T_v/C(\mathbb{R})$ is compact. In the usual topology, U is open.

For $v \in U$ and $w \in W(\mathbb{C})$ such that $f(w) = v \in V(\mathbb{R}) \subset V(\mathbb{C})$, there exists one and only one homomorphism $h(w) : \mathbb{S} \rightarrow T_v$ such that $h(w) \circ r = s_w$; after extending scalars to $\mathbb{C}$:

$$h(w) = s_w \circ z \cdot \overline{s_w} \circ \bar{z}.$$

Let $v \in U$. By the conjugacy theorem for maximal compact subgroups in $(G/C)(\mathbb{R})^0$, there exists $g \in G(\mathbb{R})$ (in fact $g \in G'(\mathbb{R})^0$) such that $gT_v g^{-1} \subset K_{\infty}$. Since T_v is a maximal torus, $gT_v g^{-1}$ contains the centre of K_{∞} . We deduce the existence of $w \in W(\mathbb{C})$ such that $f(w) = v$ and $h(w)$ is conjugate to h .

We conclude the proof of 5.1 by applying to diagram (5.1.1) and to U the following variant of Hilbert's irreducibility theorem, for which I owe the proof to J.P. Serre.

Lemma 5.1.3 Let E be a finite extension of $\mathbb{Q}$, and let a commutative diagram

$$\begin{array}{ccc} W & \xrightarrow{f} & V \\ p \downarrow & & \downarrow \\ \text{Spec}(E) & \longrightarrow & \text{Spec}(\mathbb{Q}) \end{array}$$

be given such that

- (i) V is a Zariski-open subset of an affine space over $\mathbb{Q}$;
- (ii) the geometric fibres of p are irreducible, and W is reduced;
- (iii) f is quasi-finite and dominant.

Then, for every open subset $U \subset V(\mathbb{R})$ (in the usual topology) and every extension F of E , there exists $v \in V(\mathbb{Q}) \cap U$ such that $f^{-1}(v)$ is the spectrum of an extension of E linearly disjoint from F .

Set $W' = W \otimes_E F$. The hypotheses of 4.12 are still satisfied for W'/F and $f' : W' \rightarrow W \rightarrow V$, and we need to construct $v \in V(\mathbb{Q}) \cap U$ such that $f'^{-1}(v) = f^{-1}(v) \otimes_E F$ is the spectrum of a field. This reduces us to the case $E = F$, which follows from Hilbert's irreducibility theorem (cf. Lang, Diophantine Geometry, Chap VIII).

Proposition 5.2 For $x \in M_{\mathbb{C}}(G, h)$ and K a compact open subgroup of $G(\mathbb{A}^f)$, the image of $G(\mathbb{A}^f) \cdot x \subset M_{\mathbb{C}}(G, h)$ in $M_{\mathbb{C}}(G, h)$ is dense.

This is a consequence of (0.4). The reader can check:

Proposition 5.3 Let E be a field equipped with a complex embedding ρ , X a scheme over E , and Y a reduced closed subscheme of $X_{\mathbb{C}}$. Suppose there exist finite extensions E_i of E in $\mathbb{C}$, schemes $M_{i,\alpha}/E_i$ and E_i -morphisms $v_{i,\alpha} : M_{i,\alpha} \rightarrow X \otimes_E E_i$, such that $E = \bigcap_i E_i$, that $v_i(\prod_{\alpha} M_{i,\alpha} \otimes_{E_i} \mathbb{C}) \subset Y$, and that $v_i(\prod_{\alpha} M_{i,\alpha} \otimes_{E_i} \mathbb{C})$ is dense in Y . Then Y is defined over E .

Corollary 5.4 Let $u : (G^1, h^1) \rightarrow (G^2, h^2)$ be as in 1.14. Let $M(G^1, h^1)$ and $M(G^2, h^2)$ be weakly canonical models of $M_{\mathbb{C}}(G^1, h^1)$ and $M_{\mathbb{C}}(G^2, h^2)$ over a number field E ($E(G^i, h^i) \subset E \subset \mathbb{C}$). Then $u : M(G^1, h^1) \rightarrow M(G^2, h^2)$ is defined over E (cf. 3.12).

Let $K^i \subset G^i(\mathbb{A}^f)$ be compact open subgroups such that $u(K^1) \subset K^2$. We prove that $u(K^1, K^2) : {}_{K^1}M_{\mathbb{C}}(G^1, h^1) \rightarrow {}_{K^2}M_{\mathbb{C}}(G^2, h^2)$ is defined over E . Let $u : (H, h') \rightarrow (G^1, h^1)$ be as in 3.13 and $F = E \cdot E(H, h')$. For $g \in G^1(\mathbb{A}^f)$ we have F -morphisms

$$\begin{aligned} a(g) : M_F(H, h') &\rightarrow M_F(G^1, h^1) \xrightarrow{g} M_F(G^1, h^1) \rightarrow {}_{K^1}M_F(G^1, h^1), \\ b(g) : M_F(H, h') &\rightarrow M_F(G^2, h^2) \xrightarrow{v(g)} M_F(G^2, h^2) \rightarrow {}_{K^2}M_F(G^2, h^2), \end{aligned}$$

and, after extending scalars to $\mathbb{C}$, $v(K^1, K^2) \circ a(g) = b(g)$.

By 5.1 and 5.2, we can apply 5.3 to $X = {}_{K^1}M_E(G^1, h^1) \times {}_{K^2}M_E(G^2, h^2)$, to the graph Y of v , to the fields $E_i = E \cdot E(H, h')$ for variable (H, h', u) , and to the morphisms $(a(g), b(g)) : M_{E_i}(H, h') \rightarrow X_{E_i}$: we find that the graph of v is defined over E .

In the special case $(G^1, h^1) = (G^2, h^2)$, $u = \text{Id}$, 5.4 means:

Corollary 5.5 Up to unique isomorphism, there exists at most one weakly canonical model of $M_{\mathbb{C}}(G, h)$ over E (with $E(G, h) \subset E \subset \mathbb{C}$).

Corollary 5.6 If G' is simply connected, the reciprocity law of a weakly canonical model of $M_{\mathbb{C}}(G, h)$ over E ($E(G, h) \subset E \subset \mathbb{C}$) is given by 3.9.1.

Same proof as the special case $u : (G, h) \rightarrow (T, \nu h)$ of 5.4.

Corollary 5.7 Let $v : (G^1, h^1) \hookrightarrow (G^2, h^2)$ be as in 1.15. If $M_{\mathbb{C}}(G^2, h^2)$ admits a weakly canonical model over E , with $E(G^2, h^2) \subset E(G^1, h^1) \subset E \subset \mathbb{C}$, then $M_{\mathbb{C}}(G^1, h^1)$ admits a weakly canonical model over E .

Let K^i be a compact open subgroup of $G^i(\mathbb{A}^f)$. Using 1.15, we reduce to showing that if $u(K^1) \subset K^2$ and ${}_{K^1}M_{\mathbb{C}}(G^1, h^1) \subset {}_{K^2}M_{\mathbb{C}}(G^2, h^2)$, then the closed subscheme ${}_{K^1}M_{\mathbb{C}}(G^1, h^1)$ of ${}_{K^2}M_{\mathbb{C}}(G^2, h^2) = {}_{K^2}M_E(G^2, h^2) \otimes_E \mathbb{C}$ is defined over E .

Let $u : (H, h') \rightarrow (G^1, h^1)$ be as in 3.13, and for $g \in G^1(\mathbb{A}^f)$ let $a(g)$ be the composite morphism, defined over $F = E \cdot E(H, h')$:

$$a(g) : M_F(H, h') \rightarrow M_F(G^1, h^1) \xrightarrow{v(g)} M_F(G^2, h^2) \rightarrow {}_{K^2}M_F(G^2, h^2).$$

By 5.1 and 5.2, we conclude by applying 5.3 to $X = {}_{K^2}M_E(G^2, h^2)$, to $Y = {}_{K^1}M_E(G^1, h^1)$, to the fields $E_i = E \cdot E(H, h')$ for variable (H, h', u) , and to the morphisms $a(g) : M_{E_i}(H, h') \rightarrow X_{E_i}$.

Example 5.8 For (G, h_0) as in 4.9, we construct by 5.7 and 4.21 a canonical model $M(G^0, h_0) \subset M(\text{Gp}(V, \psi), h_0) \otimes_{\mathbb{Q}} E(G^0, h_0)$.

Variant 5.9 Let (G_1, h_0) be such that, with the notations of 4.22, there exists $u : (G_1, h_0) \hookrightarrow (\text{Gp}_{2g}(F), h_0)$ (for example, (G_1^0, h_0) as in 4.13). An application of 5.7 provides a canonical model of $M(G_1, h_0)$.

Proposition 5.10 Let (G, h) be as in 3.13, let number fields E_i satisfy $E(G, h) \subset E_i \subset \mathbb{C}$, let $E = \bigcap_i E_i$, and let $M_{E_i}(G, h)$ be a weakly canonical model of $M_{\mathbb{C}}(G, h)$ over E_i . Then there exists a unique model $M_E(G, h)$ of $M_{\mathbb{C}}(G, h)$ over E such that for every i , the model $M_E(G, h) \otimes_E E_i$ over E_i is isomorphic to $M_{E_i}(G, h)$.

Given the uniqueness statement, we may assume there are finitely many E_i . Let F be the compositum of the E_i in $\mathbb{C}$, and let $\mathfrak{F}$ be the set of subextensions of F containing one of the E_i . Using 5.5, we obtain for every $F' \in \mathfrak{F}$ a model $M_{F'}(G, h)$, and for every inclusion $F' \subset F''$ ($F', F'' \in \mathfrak{F}$) an isomorphism of models $\alpha(F'', F') : M_{F'}(G, h) \otimes_{F'} F'' \xrightarrow{\sim} M_{F''}(G, h)$.

Let $u : (H, h') \rightarrow (G, h)$ be as in 3.13, with $E(H, h')$ linearly disjoint from F over $E(G, h)$ (5.1). Set $E_1 = E(H, h') \cdot E$ and $F'_1 = E_1 \cdot F' = E(H, h') \cdot F'$ for $F' \in \mathfrak{F}$. Let $g \in G(\mathbb{A}^f)$ and $K \subset G(\mathbb{A}^f)$

be a compact open subgroup. Since $M_{F'}(G, h)$ is weakly canonical, we have an F'_1 -morphism

$$a'(g) : M_{F'_1}(H, h') \rightarrow M_{F'_1}(G, h) \xrightarrow{g} M_{F'_1}(G, h) \rightarrow {}_K M_{F'_1}(G, h).$$

Because $F'_1 = E_1 \otimes_E F'$, this can be "interpreted" as a morphism of F' -schemes

$$a(g) : M_{E_1}(H, h') \otimes_E F' \rightarrow {}_K M_{F'}(G, h).$$

The $a(g)$ satisfy $\alpha(F'', F')a(g)_{F''} = a(g)$. We conclude by 5.2 and the following lemma, applied to $X^1 = \coprod_{g \in G(\mathbb{A}^f)} M_{E_1}(H, h)$. Its proof is left as an exercise to the reader.

Lemma 5.10.1 Let F/E be a finite field extension and $\mathfrak{F}$ a set of subfields of F whose intersection is E , such that

$$(F' \in \mathfrak{F} \text{ and } F' \subset F'' \subset F) \implies F'' \in \mathfrak{F}.$$

- (i) The functor which associates to a scheme X/E the system $X_{\mathfrak{F}}$ of schemes $X_{F'}/F'$ ($F' \in \mathfrak{F}$) and the isomorphisms $X_{F'} \otimes_{F'} F'' \xrightarrow{\sim} X_{F''}$ ($F' \subset F''$, $F', F'' \in \mathfrak{F}$) is fully faithful.
- (ii) For a system $\mathcal{X} = \{X(F')/F' \text{ } (F' \in \mathfrak{F}); \alpha(F'', F') : X(F') \otimes_{F'} F'' \xrightarrow{\sim} X(F'') \text{ } (F' \subset F'', F', F'' \in \mathfrak{F})\}$ to come from a scheme X/E , it suffices that the $X(F')/F'$ be quasi-projective and that there exist a scheme X^1/E and a morphism $u : X^1_{\mathfrak{F}} \rightarrow \mathcal{X}$ such that the morphisms $u(F') : X^1_{F'} \rightarrow X(F')$ have dense image.

Remark 5.10.2 Under the hypotheses of 5.10, for $u : (H, h') \hookrightarrow (G, h)$ as in 3.13, the field of definition of the morphism $u : M(H, h') \rightarrow M(G, h)$ is contained in the extension $\bigcap_i E(H, h') \cdot E_i$ of $E(H, h') \cdot E$.

Proposition 5.11 Let $h : \mathbb{S} \rightarrow G_{\mathbb{R}}$ be as in 3.13 and $\delta : \mathbb{S} \rightarrow C_{\mathbb{R}}^0$ (where C is the centre of G). If $E(G, h), E(C, \delta) \subset E \subset \mathbb{C}$ and if $M(G, h)$ admits a weakly canonical model over E , then $M(G, h \cdot \delta)$ admits one as well.

Let K be a compact open subgroup of $G(\mathbb{A}^f)$ and $L = K \cap C^0(\mathbb{A}^f)$. Over $\mathbb{C}$, the morphism

$$u : (G \times C^0, (h, \delta)) \rightarrow (G, h\delta), \quad (g, c) \mapsto gc,$$

induces morphisms

$$u_K : {}_K M_E(G, h) \times {}_L M_{\mathbb{C}}(C^0, \delta) \rightarrow {}_K M_{\mathbb{C}}(G, h\delta).$$

Let $d : C^0 \rightarrow G \times C^0$ be the morphism $c \mapsto (c, c^{-1})$. The subgroup $d(C^0(\mathbb{A}^f))$ of $G \times C^0(\mathbb{A}^f)$ acts on $M_{\mathbb{C}}(G, h) \times M_{\mathbb{C}}(C^0, \delta)$ via the quotient $\pi_0(C^0(\mathbb{A}))$ of $C^0(\mathbb{A}^f)$, and u_K induces an isomorphism

$$\overline{u_K} : ({}_K M_E(G, h) \times {}_L M_{\mathbb{C}}(C^0, \delta)) / (L \backslash \pi_0(C^0(\mathbb{A}))) \xrightarrow{\sim} {}_K M_{\mathbb{C}}(G, h\delta).$$

We define ${}_K M_E(G, h\delta)$ as the quotient

$${}_K M_E(G, \delta) \times {}_L M_E(C^0, \delta) / (L \backslash \pi_0(C^0(\mathbb{A})))$$

and verify that we obtain a weakly canonical model. By construction, we have

$$(5.11.1) \quad M_E(G, h) \times_E M_E(C^0, \delta) \rightarrow M_E(G, h\delta).$$

Conjecture 5.12 For every (G, h_0) as in 3.13, there exists a canonical model of $M_{\mathbb{C}}(G, h_0)$.

Suppose, as a typical case, that G is $\mathbb{Q}$ -simple and adjoint. Let $\tilde{G}$ be the universal covering of G and w a weight of $\tilde{G}_{\mathbb{C}}$. If, with the notations of 3.7, the conjugacy class of h_0r is described by integers $(n_i)_{i \in |\Delta|}$, and $w = \sum m_i \alpha_i$, set $\langle w, h_0r \rangle = \sum n_i m_i$. So far, one has been able to attack 5.12 only when $\tilde{G}$ admits a non-trivial representation V such that every irreducible component V' of $V_{\mathbb{C}}$ satisfies: (*) as w runs over the weights of V' , $\langle w, h_0r \rangle$ takes at most two values.

Let σ be the opposition involution. Condition (*) also means that the dominant weight w of V' satisfies $\langle w, h_0r \rangle + \langle \sigma w, h_0r \rangle = 0$ or 1 . No such representation V exists if G is exceptional (the non-compact factors of $G_{\mathbb{R}}$ are then of type $E_{6(-14)}$ or $E_{7(-25)}$), or of triality D_4 type, nor if G is of type D_n and $G_{\mathbb{R}}$ has non-compact factors of both types $D_n^{\mathbb{H}}$ and $D_n^{\mathbb{R},2}$.

Chapter 6

Strange models

6.1

Let F be a totally real number field, and let

$$(6.1.1) \quad {}_F\Sigma : 0 \rightarrow {}_FG' \rightarrow {}_FG \rightarrow \mathbb{G}_m \rightarrow 0$$

be a form over F of the exact sequence

$$(6.1.2) \quad {}_F\Sigma_0 : 0 \rightarrow \mathrm{Sp}_{2n} \rightarrow \mathrm{Gp}_{2n} \xrightarrow{\mu} \mathbb{G}_m \rightarrow 0,$$

where Gp denotes a group of symplectic similitudes. We assume that, for each real place τ of F , $\Sigma \otimes_{F,\tau} \mathbb{R}$ is of one of the following types:

- (a) ${}_F\Sigma \otimes_{F,\tau} \mathbb{R}$ is isomorphic to ${}_F\Sigma_0 \otimes_{F,\tau} \mathbb{R}$, or
- (b) $G' \otimes_{F,\tau} \mathbb{R}$ is compact.

Denote by $\tau_1, \dots, \tau_g$ the real places of F , and assume that the places of type (a) are $\tau_1, \dots, \tau_r$ with $r > 0$.

6.2

The group ${}_FG$ can be described as follows:

- (a) Choose a quaternion algebra B over F , indefinite at the places τ_i for $i \leq r$, and definite for $i > r$. Denote by $x \mapsto \bar{x}$ its canonical involution.
- (b) Choose a free B -module V , equipped with a non-degenerate F -bilinear form Φ , symmetric, and such that

$$\Phi(bx, y) = \Phi(x, \bar{b}y).$$

- (c) Assume that, for $i > r$, the form obtained from Φ by base change $\tau_i : F \rightarrow \mathbb{R}$ is positive definite.

- (d) Take ${}_FG$ to be the group of B -linear similitudes of Φ ; in other words, the elements $g \in G(\mathbb{Q})$ are those $g \in \mathrm{GL}_B(V)$ for which there exists $\mu(g) \in F^*$ satisfying

$$\Phi(gx, gy) = \mu(g)\Phi(x, y).$$

6.3

Denote by

$$\Sigma : 0 \rightarrow G' \rightarrow G \rightarrow F^* \rightarrow 0$$

the exact sequence of algebraic groups over $\mathbb{Q}$ obtained from ${}_F\Sigma$ by Weil restriction of scalars from F to $\mathbb{Q}$. We have

$$(6.3.1) \quad G_{\mathbb{R}} = \prod_{\tau} {}_FG \otimes_{F, \tau} \mathbb{R} \quad (\tau \text{ real place}).$$

A homomorphism $h : \mathbb{S} \rightarrow G_{\mathbb{R}}$ is defined by its coordinates $h_i : \mathbb{S} \rightarrow {}_FG \otimes_{F, \tau_i} \mathbb{R}$. By 1.6, there exists one and only one conjugacy class of homomorphisms h such that, for $i \leq r$, h_i is of type 1.6, and for $i > r$, h_i is trivial. Let h_0 be a representative of this class.

The field $E(G, h_0)$ is the totally real subfield of $\mathbb{C}$ generated by the sums $\sum_{i=1}^r \tau_i(f)$ for $f \in F$.

Theorem 6.4 With the above notations, $M_{\mathbb{C}}(G, h_0)$ admits a canonical model $M(G, h_0)$.

Let Z be a totally imaginary quadratic extension of F . Set $V = V \otimes_F Z$, $L = B \otimes_F Z$, and let G' be the algebraic subgroup of $\mathrm{GL}_L(V)$ generated by G and Z^* :

$$(6.4.1) \quad 0 \rightarrow F^* \xrightarrow{(f, f^{-1})} G \times Z^* \xrightarrow{q} G' \rightarrow 0.$$

Let $x \mapsto \bar{x}$ be the involution of L obtained as the tensor product of the involutions of B and Z . Up to a factor in F^* , there exists a unique F -bilinear alternating form ψ_Z on Z . If ${}_F\psi$ is the F -bilinear alternating form on V given by the tensor product of Φ and ψ_Z , then

$${}_F\psi(\ell x, y) = {}_F\psi(x, \bar{\ell} y).$$

Set $\psi(x, y) = \mathrm{Tr}_{F/\mathbb{Q}} \psi(x, y)$.

For each real place of F , choose a complex embedding of Z that induces it. Then we have $Z^*(\mathbb{R}) \simeq \prod_1^g \mathbb{C}^*$. Let h_Z be the homomorphism $\mathbb{S} \rightarrow Z_{\mathbb{R}}^*$ with coordinates $h_i : \mathbb{S}(\mathbb{R}) = \mathbb{C}^* \rightarrow \mathbb{C}^*$ ($1 \leq i \leq g$) equal to $z \mapsto 1$ for $i \leq r$, and to $z \mapsto z^{-1}$ for $i > r$.

For each $h : \mathbb{S} \rightarrow G_{\mathbb{R}}$ as in 6.3, set $h' = q \circ (h \times h_Z) : \mathbb{S} \rightarrow G'_{\mathbb{R}}$.

Lemma 6.4.2 There exists $b \in L$ such that, for a suitable choice of ψ_Z , the form

$$\psi(x, bh'_0(i)y) \quad \text{on} \quad V \otimes_{\mathbb{Q}} \mathbb{R}$$

is symmetric and positive definite.

This can be verified after extending scalars from $\mathbb{Q}$ to $\mathbb{R}$.

This lemma and 5.9 allow us to construct a canonical model of $M(G', h'_0)$ (because G' is contained in the group of symplectic similitudes of ${}_F\psi(x, by)$). Applying 5.11 and 5.7, we obtain a weakly canonical model of $M_{\mathbb{C}}(G, h_0)$ over the extension $E(Z^*, h_Z)$ of $E(G, h_0)$. We conclude by 5.10, 5.10.2, and the following lemma, which can be deduced from 5.13.

Lemma 6.5 Let F be a totally real number field equipped with a set S of real embeddings. Let

F^* be the number field corresponding to the subgroup of $\text{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$ that stabilises S . For Z a totally imaginary quadratic extension of F , equipped with a set T of complex embeddings, one above each element of S , let similarly $Z^* \supset F^*$ be the field of invariants of the stabiliser of T . Then, for every extension E of F^* , there exist Z and T such that Z^* is linearly disjoint from E .

Remark 6.6 The canonical map from 1.13, 1.14

$$(6.6.1) \quad M_{\mathbb{C}}(G, h_0) \times M_{\mathbb{C}}(Z^*, h_Z) \longrightarrow M_{\mathbb{C}}(G', h'_0)$$

can be interpreted as follows. Each conjugate of h defines on $V_{\mathbb{C}}$ a Hodge bigradation invariant under F :

$$V_{\mathbb{C}} = V^{-1,0} \oplus V^{0,-1} \oplus V^{0,0} \quad (\overline{V^{pq}} = V^{qp}).$$

Note that the weight grading $V_{\mathbb{C}} = \bigoplus_n V^n$ with $V^n = \sum_{p+q=n} V^{pq}$ is not defined over $\mathbb{Q}$ if $r \neq g$.

Similarly, $Z_{\mathbb{C}}$ is bigraded:

$$Z_{\mathbb{C}} = Z^{-1,0} \oplus Z^{0,-1} \oplus Z^{0,0}.$$

The miracle is that the $\mathbb{Q}$ -vector space $V^{\cdot} = V \otimes_F Z$, equipped with the tensor product Hodge bigradation

$$V_{\mathbb{C}}^{\cdot} = (V_{\mathbb{C}}^{\cdot})^{-1,0} \oplus (V_{\mathbb{C}}^{\cdot})^{0,-1} \quad (!)$$

is the H_1 of an abelian variety.

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